

Tan Dai Ngo

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EDUCATION

University of Victoria | Master of Engineering, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Courses:** Optimization for ML, Artificial Intelligence, Data Mining, Information Theory. **GPA:** 8.78 / 9.00.

University of Chicago | Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Courses:** Algorithms, Distributed Systems, Databases, Applied Software Engineering, Networks, Web Development.

University of Washington | Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot, keeping Cassandra query response times under 3 seconds.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Debugged production issues with Splunk and built a Mobileum email notifier over the Microsoft Graph API.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, PyTorch, XGBoost *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows under split first preprocessing; binary XGBoost reached 99.50% recall at 0.938% FPR.
- Designed a four condition holdout that exposed session leakage, dropping PR-AUC from 0.981 to 0.125.
- Scripted an integrity check rebuilding the 60 second connection count on a truncated capture, proving causality.

Ocean Data ML Platform with RAG | Python, Spark, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10.8M NOAA and Ocean Networks Canada observations into Bronze and Spark Silver layers, then built a 3.69M row Gold feature table for forecasting.
- Served model inference and plain English retrieval behind FastAPI with Qdrant and Ollama, at hit@k 0.90.
- Tuned XGBoost with Optuna and tracked runs in MLflow, and in a separate benchmark beat persistence on air temperature by 48% RMSE at 12 hours, with a Streamlit dashboard.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Built an automated Coinbase trading bot in Python that buys only when the 8 period EMA is above the 20 period EMA, price is above the 200 period EMA, and the fee adjusted return is positive.
- Implemented a mean reversion engine that buys a 1% dip inside a rolling 60 minute window and sells on a 2% rise, with per market thresholds, cooldowns and open trade caps in a CSV config.
- Deployed it on AWS EC2 to run unattended, adding a Lambda and SNS service that alerted on volume spikes.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, HTML, CSS, JavaScript, MATLAB, Cypher

Machine Learning & AI: PyTorch, XGBoost, scikit-learn, pandas, NumPy, DistilBERT, Optuna, MLflow, RAG, Ollama

Data & Backend: Spark, PySpark, ETL, Kafka, RabbitMQ, Cassandra, MySQL, MongoDB, Qdrant, FastAPI, Spring Boot

Platform & Tools: Docker, AWS (EC2, S3), Jenkins CI/CD, Git, GitHub, Bitbucket, Postman, Splunk, pytest, Jira

Certifications & Competitions: Google Advanced Data Analytics; Shell.ai Hackathon 2025; IMC 64 Bids, top 250 of 1,500